

First-In First-Out (FIFO) 64x4 Serial Memory

57401/67401

Features/Benefits

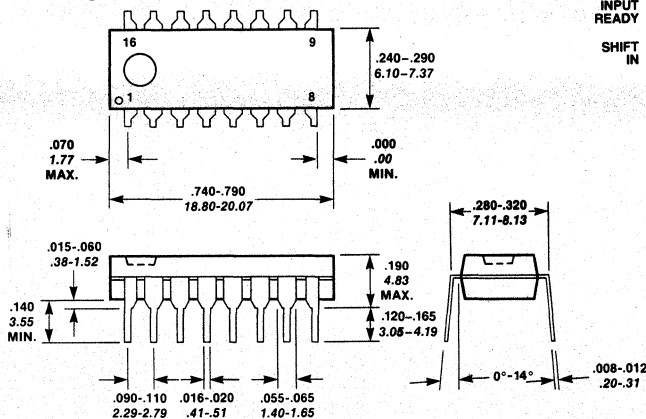
- 12 MHz shift in, shift out typical rates
- Advanced Schottky bipolar processing
- TTL inputs and outputs
- Readily expandable in word and bit dimensions
- Asynchronous or synchronous operation
- Pin compatible with Fairchild's F3341 MOS FIFO and ten times as fast

Description

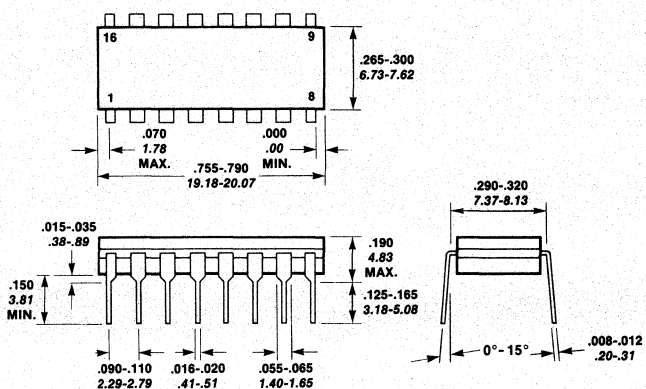
The 57401 is an expandable "fall-through" high speed First-In First-Out (FIFO) memory organized 64 words by four bits. A 15 MHz data rate allows usage in high speed tape or disc controllers and communication buffer applications. The 67401 is specified over the commercial operating range 0–75°C and the 57401 is specified over the military operating range of –55°C to +125°C.

Package Drawing

N16 Plastic DIP



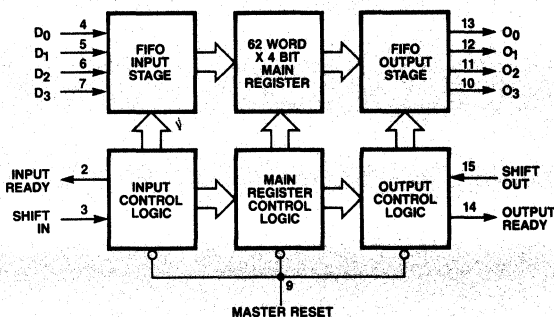
J16 Ceramic DIP



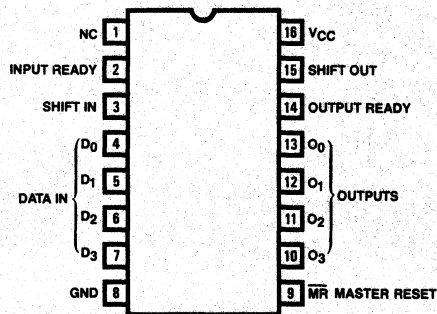
Selection Guide

PART NUMBER	PACKAGE	TEMPERATURE RANGE
57401	J16	Military
67401	N16	Commercial

Block Diagram



Pin Configuration



UNLESS OTHERWISE SPECIFIED:
ALL DIMENSIONS MIN.-MAX. IN INCHES.
ALL DIMENSIONS MIN.-MAX. IN MILLIMETERS.

Absolute Maximum Ratings

Supply voltage V_{CC}	Operating 7V/12V
Input voltage	5.5V/12V
Off-state output voltage	5.5V/12V
Storage temperature range	-65°C to 150°C

Recommended Operating Conditions

SYMBOL	PARAMETER	MILITARY			COMMERCIAL			UNIT
		MIN	NOM	MAX	MIN	NOM	MAX	
V_{CC}	Supply voltage	4.5	5.0	5.5	4.75	5.00	5.25	V
I_{OH}	High level output current			0.9			0.9	mA
I_{OL}	Low level output current			8			8	mA
T_A	Operating free air temperature	-55		125	0		75	°C

Electrical Characteristics

Over Recommended Operating Free Air Temperature

SYMBOL	PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
V_{IH}	High-level input voltage			2			V
V_{IL}	Low-level input voltage					0.8	V
V_{IC}	Input clamp voltage	$V_{CC} = \text{MIN}, I_I = -18\text{mA}$				-1.5	V
V_{OL}	Low-level output voltage	$V_{CC} = \text{MIN}, V_{IH} = 2\text{V}$ $V_{IL} = 0.8\text{V}, I_{OL} = \text{MAX}$				0.5	V
I_I	Input current at maximum input voltage	$V_{CC} = \text{MAX}, V_I = 5.5\text{V}$				1	mA
I_{IH}	High-level input current	$V_{CC} = \text{MAX}, V_I = 2.4\text{V}$				50	μA
I_{IL1}	Low-level input current inputs D ₀ –D ₃ , MR	$V_{CC} = \text{MAX}, V_F = 0.45\text{V}$				-0.8	mA
I_{IL2}	Low-level input current inputs S ₁ , S ₀	$V_{CC} = \text{MAX}, V_F = 0.45\text{V}$				-1.6	mA
C_I	Input capacitance	$V_{CC} = 5.0\text{V}$	$V_I = 2.0\text{V}$		7		pF
C_O	Output capacitance	$f = 1\text{ MHz}, T_A = 25^\circ\text{C}$	$V_O = 2.0\text{V}$		8		
I_{CC}	Supply current	$V_{CC} = \text{MAX}^1$			111	150	mA

THREE STATE OUTPUT

V_{OH}	High-level output voltage	$V_{CC} = \text{MIN}, V_{IH} = 2\text{V}$ $V_{IL} = 0.8\text{V}, I_{OH} = \text{MAX}$	2.4	V
I_{OZH}	Off-state output current high-level voltage applied	$V_{CC} = \text{MAX}, V_{IH} = 2\text{V}$ $V_O = 2.4\text{V}$		100 μA
I_{OZL}	Off-state output current low-level voltage applied	$V_{CC} = \text{MAX}, V_{IH} = 2\text{V}$ $V_O = 0.4\text{V}$		-100 μA
I_{OS}	Short-circuit output current	$V_{CC} = 6\text{V}, V_O = .5\text{V}$	-20	-90 mA

NOTE: 1. I_{CC} measured with worst case sequence; FIFO reset, one shift-in cycle with data inputs held low, measure I_{CC} with all inputs held low.

Switching Characteristics

See Figure 1

SYMBOL	PARAMETER	FIGURE	57401—MILITARY 5 ± 10%, -55°C to +125°C			67401—COMMERCIAL 5 ± 5%, 0°C to 70°C		
			MIN	MAX	UNIT	MIN	MAX	UNIT
t _{IN}	Shift in Clock Rate	1	7		MHz	10		MHz
t _{SIH}	Shift in High Time	1	45		ns	35		ns
t _{SIL}	Shift in Low Time	1	45		ns	35		ns
t _{IRL}	Input Ready ON Delay	1		60	ns		45	ns
t _{IRH}	Input Ready OFF Delay	1		60	ns		45	ns
t _{IDS}	Input Data Set Up	1	10		ns	5		ns
t _{IDH}	Input Data Hold Time	1	55		ns	45		ns
f _{OUT}	Shift Out Clock Rate	2	7		MHz	10		MHz
t _{SOH}	Shift Out High Time	2	45		ns	35		ns
t _{SOL}	Shift Out Low Time	2	45		ns	35		ns
t _{ORL}	Output Ready ON Delay	2		65	ns		55	ns
t _{ORH}	Output Ready OFF Delay	2		65	ns		55	ns
t _{ODH}	Output Data Hold Time	2	10		ns	10		ns
t _{ODS}	Output Data Delay	2		65	ns		55	ns
t _{PT}	Data Throughput Time	Note 1		4.0	μs		3.0	μs
t _{MRW}	Master Reset Pulse	Note 2	30		ns	25		ns
t _{MRORL}	Master Reset to OR Low	5		65	ns		55	ns
t _{MRIRH}	Master Reset to IR High	5		45	ns		35	ns
t _{MRS}	Master Reset to SI	5	45		ns	35		ns
t _{IPH}	Input Ready Pulse High	4	45		ns	35		ns
t _{IPL}	Input Ready Pulse Low	6	45		ns	35		ns
t _{OPH}	Output Ready Pulse High	3	45		ns	35		ns
t _{OPL}	Output Ready Pulse Low	7	45		ns	35		ns

NOTES: 1. This parameter defines total time from the time data is present at D₀-D₃ to the time it is available at O₀-O₃ with FIFO initially empty and total time from the time data is extracted from O₀-O₃ to IR High with FIFO initially full.

2. Master Reset clears the 57401/67401 to the all cells empty state.

Functional Description

Data Input

Data is entered into the FIFO on D₀-D₃ inputs. To enter data the Input Ready (IR) output should be HIGH, indicating that the first location is ready to accept data. Data then present at the four data inputs is entered into the first location when the Shift In (SI) is brought HIGH. A SI HIGH signal causes the RI to go LOW. Data remains at the first location until SI is brought LOW. When SI is brought LOW and the FIFO is not full, IR will go HIGH, indicating that more room is available. Simultaneously, data will propagate to the second location and continue shifting until it reaches the output stage or a full location. If the memory is full, IR will remain LOW.

Data Transfer

Once data is entered into the second cell, the transfer of any full cell to the adjacent (downstream) empty cell is automatic, activated by an on-chip control. Thus data will stack up at the

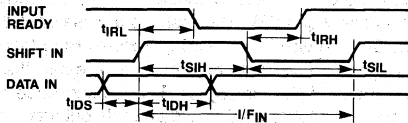
end of the device while empty locations will "bubble" to the front. t_{PT} defines the time required for the first data to travel from input to the output of a previously empty device.

Data Output

Data is read from the O₀-O₃ outputs. When data is shifted to the output stage, Output Ready (OR) goes HIGH, indicating the presence of valid data. When the OR is HIGH, data may be shifted out by bringing the Shift Out (SO) HIGH. A HIGH signal at SO causes the OR to go LOW. Valid data is maintained while the SO is HIGH. When SO is brought LOW the upstream data, provided that stage has valid data, is shifted to the output stage. When new valid data is shifted to the output stage, OR goes HIGH. If the memory is emptied, OR stays LOW.

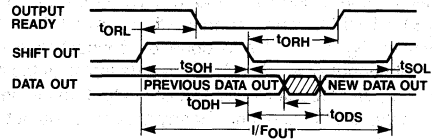
Input Ready and Output Ready may also be used as status signals indicating that the FIFO is completely full (Input Ready stays LOW for at least t_{PT}) or completely empty (Output Ready stays LOW for at least t_{PT}).

Timing Diagrams



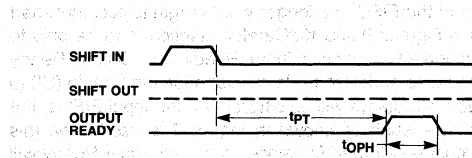
1. Input data must remain stable during t_{IDS} and t_{IDH} .
2. Input Ready HIGH indicates that space is available and a Shift In pulse may be applied. Input Ready LOW indicates that the FIFO is full or that a previous Shift In operation is not complete. Shift In pulses applied while Input Ready is LOW will be ignored.
3. The rise of Input Ready indicates that the data at the D0-D3 inputs has been accepted and that the input stage is empty.

Figure 1. FIFO Input Timing



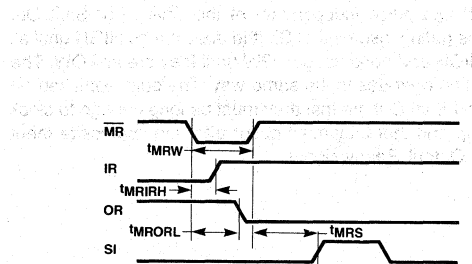
1. Output data will be stable at the rise of Output Ready.
2. Output Ready HIGH indicates that data is available and a Shift Out pulse may be applied. Output Ready LOW indicates that the FIFO is empty or that a previous Shift Out operation is still in progress. Shift Out pulses applied while Shift Out Ready is LOW will be ignored.
3. The rise of Output Ready indicates that new data has been loaded into the output stage.

Figure 2. FIFO Output Timing



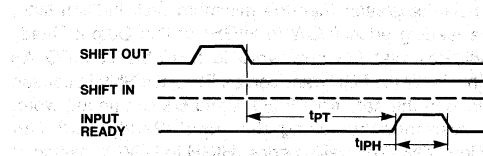
1. FIFO initially empty.
2. Shift Out held high (as shown).

Figure 3. t_{OPH} Specification



1. FIFO initially full.

Figure 4. t_{TPH} Specification



1. FIFO initially full.
2. Shift In held high (as shown).

Figure 5. FIFO Master Reset Timing

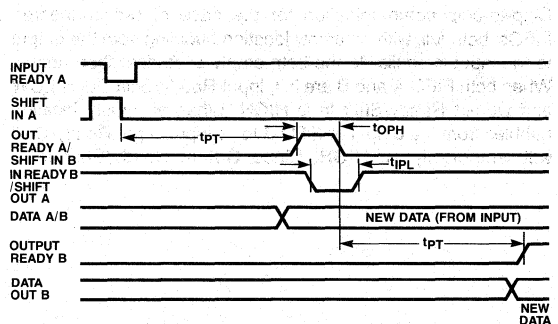
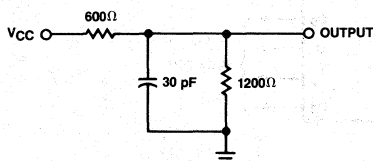


Figure 6. FIFO/FIFO Communication: Empty/Input Timing

Standard Test Circuit



Input Pulse Amplitude = 2.5V
Input Rise and Fall Time = 5.0 ns from 1.0V to 2.0V
Measurements made at 1.50V

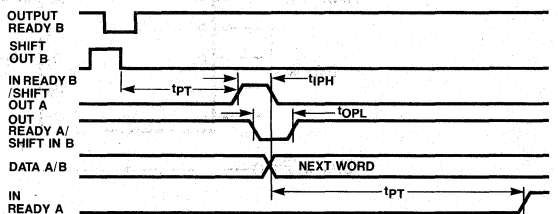


Figure 7. FIFO/FIFO Communication: Full/Output Timing

Expansion

FIFOs can be connected serially to form larger effective FIFOs: i.e., two 64 x 4 FIFOs can be connected to form a 128 x 4 FIFO, as shown in Figure 8. This is done by connecting the data outputs of the input side FIFO, A, to the data inputs of the output side FIFO, B, and Output Ready and Shift Out Pins of the input FIFO, A, to the Shift In and Input Ready pins of the output FIFO, B, respectively.

If both FIFOs are empty, the Input Ready/Shift Out pins between the FIFOs will be HIGH and the Output Ready/Shift In pins between the FIFOs will be LOW. If the Shift In pin of FIFO A is clocked, the data will propagate to the output. This will set the output full indicator, causing the Output Ready/Shift In combination to rise temporarily. However, since the Input Ready/Shift Out combination is HIGH this enables the Shift Out clock and the full indicator is reset immediately after the rise of Output Ready/Shift In. The result is a positive going pulse on the Output Ready/Shift In line which is t_{OPH} , Output Pulse High (empty), as shown in Figure 6. The FIFO chip is designed such that the pulse is greater than the minimum Shift In High time, t_{SIH} . The leading edge (LOW to HIGH) of this Output Ready pulse indicates that the new word is available at FIFO A's output. This LOW to HIGH transition on FIFO B's Shift In causes FIFO B to load the data from FIFO A into B's first (input) word. FIFO B responds by taking the Input Ready/Shift Out combination LOW. The trailing edge, HIGH to LOW transition of Output Ready/Shift In pulse allows the word in FIFO B to propagate on to its output. Once propagation has begun, FIFO B will again raise Input Pulse Low (empty). The FIFO is designed such that this pulse is greater than the minimum Shift Out Low time, t_{SOL} .

Chip-to-chip communication for the case of two connected FIFOs, both full, with an empty location bubbling from the output to the input is similar to the both-empty case described above. When both FIFO A and B are full, Input Ready/Shift Out is LOW and Output Ready/Shift In is HIGH. When an empty location bubbles from the output of FIFO B to its input, Input Ready/Shift will temporarily go HIGH. Since Output Ready/Shift In is

already HIGH, the Shift In clock is enabled and Input Ready/Shift Out will immediately go LOW. The result is a pulse on the Input Ready/Shift Out line which is t_{IPH} , Input Pulse High (full). The FIFO is designed such that this pulse is greater than the minimum Shift Out High time, t_{SOH} . The leading edge of this pulse will cause FIFO A to load new data at its outputs, and the trailing edge will cause the newly created empty location to bubble through FIFO A. The rise of the Input Ready/Shift Out pulse will cause Output Ready/Shift In to go LOW. Output Ready/Shift In will go HIGH again when new data has been loaded into FIFO A's output. This creates a negative pulse on the Output Ready/Shift In line which is t_{OPL} , Output Pulse Low (full). The FIFO is designed such that t_{OPL} is greater than the minimum Shift Out Low time.

Word Length Expansion

Expansion of the FIFO to a longer word length is accomplished as shown in Figures 9 and 10. Careful attention must be paid to the generation of composite Input Ready and Output Ready signals. Because of device-to-device variation the Shift In (SI) to Input Ready (IR) delays will be different on the Input FIFOs. If a simple scheme such as shown in Figure 9 is used, then this variation must be allowed for, since composite Input Ready will go LOW as soon as the fastest IR goes LOW, but Shift In cannot go LOW until all three IRs for the input FIFOs are LOW. The user must provide sufficient delay to insure this.

Figure 9 shows one way to eliminate any potential timing problems. The gates and flip-flops are used to generate a Shift In or Shift Out pulse independent of the Shift In or Shift Out signal. The gating insures that Shift In does not go HIGH until all IRs are HIGH and does not go LOW until they are all LOW. The output gating operates in the same way. The only restriction on Shift In and Shift Out are that they must be long enough to clock the flip-flop and that they must occur after the composite Input Ready or Output Ready signal.

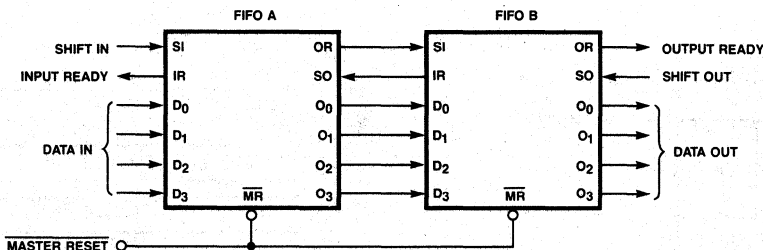


Figure 8. 128 x 4 FIFO

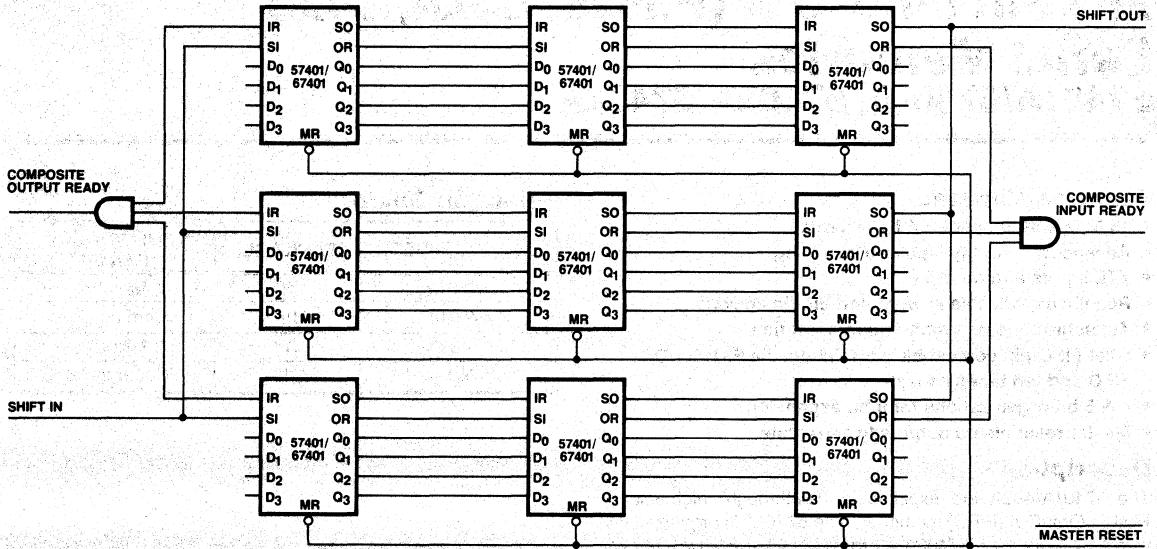


Figure 9. 192 x 12 FIFO—Unlatched Control Lines

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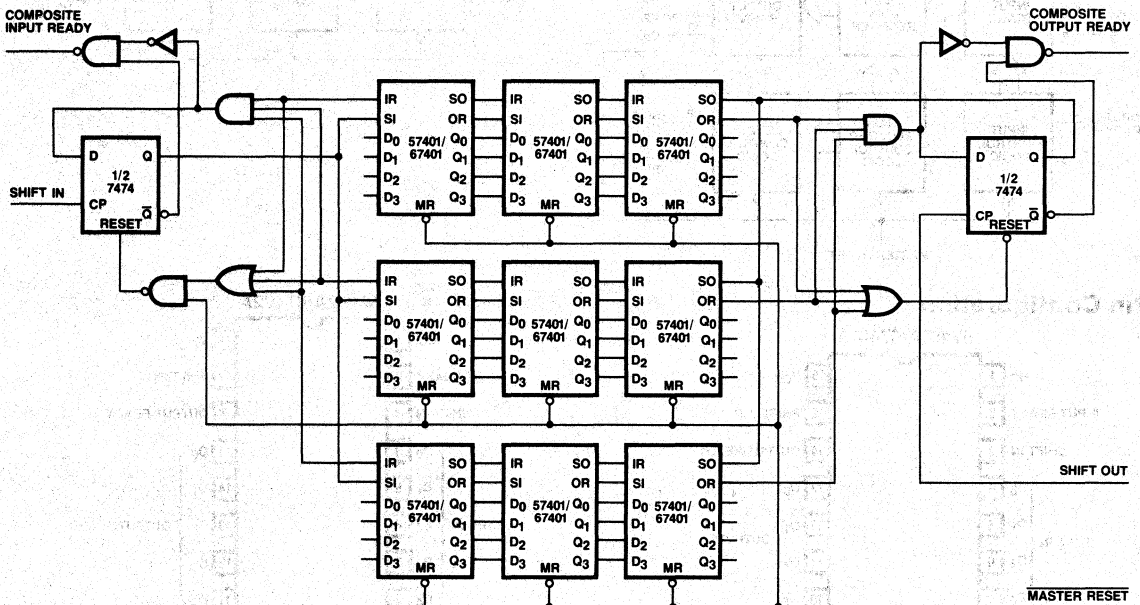


Figure 10. 192 x 12 FIFO—Latched Control Lines